

Customer Retention & Churn Analysis

Internship Project – Future Interns

Repository Name: FUTURE_DS_02

Prepared by: Sravani Korlepara

Tools Used: Microsoft Power BI, Microsoft Excel

Dataset: Telco Customer Churn Dataset

1. Project Overview

Customer retention is one of the most important factors influencing the growth and profitability of subscription-based businesses. Acquiring new customers is generally more expensive than retaining existing ones. This project analyzes customer churn behavior using the Telco Customer Churn dataset to identify retention trends, customer lifetime patterns, and the factors contributing to customer churn.

The dashboard was developed using Microsoft Power BI to provide business stakeholders with interactive insights for improving customer retention strategies.

2. Project Objectives

The objectives of this project are:

- Analyze customer churn patterns.
- Measure customer retention using key performance indicators.
- Identify customer segments with higher churn rates.
- Understand customer lifetime through tenure analysis.
- Recommend business strategies to improve retention.

3. Dataset Description

The project uses the Telco Customer Churn Dataset containing customer demographic information, subscription details, payment methods, service usage, monthly charges, tenure, and churn status.

The dataset includes approximately **7,043 customer records** and multiple attributes describing customer behavior.

4. Data Cleaning

The following preprocessing steps were performed before analysis:

- Checked for missing values.
- Verified and corrected data types.
- Converted Monthly Charges and Total Charges into Currency format.
- Removed unnecessary blank values.
- Validated customer records for consistency.
- Prepared the cleaned dataset for Power BI visualization.

5. Dashboard Overview

The Power BI dashboard provides an interactive overview of customer retention and churn.

Key Performance Indicators (KPIs)

- Total Customers
- Churned Customers
- Churn Rate
- Average Customer Tenure

Interactive Filters

- Gender
- Contract Type
- Internet Service

- Payment Method

Visualizations

- Churn by Contract Type
- Customer Distribution by Internet Service
- Churn by Payment Method
- Average Monthly Charges by Churn
- Average Tenure by Churn

6. Key Insights

The dashboard analysis revealed the following:

- Customers with month-to-month contracts exhibit significantly higher churn than those on long-term contracts.
- Customers with shorter tenure are more likely to leave the service.
- Higher monthly charges are associated with increased churn.
- Certain payment methods show relatively higher churn rates.
- Internet service type influences customer retention behavior.

7. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Encourage customers to upgrade from month-to-month plans to annual or multi-year contracts.
- Provide personalized retention offers for customers with shorter tenure.
- Improve customer support for high-risk customer segments.
- Introduce loyalty rewards and discounts for long-term subscribers.
- Review pricing strategies for customers with high monthly charges.
- Monitor churn trends regularly using the dashboard.

8. Business Impact

Implementing these recommendations can help businesses:

- Reduce customer churn.
- Increase customer lifetime value.
- Improve customer satisfaction.
- Increase recurring subscription revenue.
- Support data-driven decision making.

9. Conclusion

This project successfully analyzed customer retention and churn behavior using Microsoft Power BI. Interactive visualizations and KPIs provided valuable insights into customer behavior, enabling business stakeholders to identify churn patterns and develop effective retention strategies. The dashboard serves as a practical decision-support tool for subscription-based businesses aiming to improve long-term customer relationships.

10. Skills Gained

- Customer Retention Analysis
- Churn Analysis
- Data Cleaning
- KPI Development
- Interactive Dashboard Design
- Business Intelligence
- Microsoft Power BI
- Data Visualization
- Business Insight Generation

Appendix

Tools Used

- Microsoft Power BI
- Microsoft Excel

Dataset

Telco Customer Churn Dataset